PMI-ACP®
EXAM PREP
PMI Authorized Training Partner
BOOTCAMP
Session 7

- This course will assist learners in preparing for PMI's PMI-ACP Exam (2024 Update)

ATTENDENCE TRACKING

Percipio Users:
Name is based on your log in information in Percipio

Using Zoom:
Enter your first and last name

BREAKS

Yes! We will have periodic breaks

For attendance purposes, please stay logged in during all breaks.

CHAT vs Q&A

Please use the **Chat** for:

- **Greetings** before the session starts and during breaks
- Once the session starts , the chat may be closed or changed to *Hosts & Panelists Only* to minimize disruptions and focus on important information.
- The instructor may open the chat during the session for student **to respond to the instructor's questions** and create a group dialog.

CHAT vs Q&A

Please use the **Q&A** for:

- **Technical assistance** – Begin with: Percipio or Non-Percipio student
- Guidance on how to **access course material** – Begin with: Percipio or Non-Percipio
- Clarification and **questions on lecture points**, if not answered by instructor
- The Q&A may be open and closed throughout the session to allow us to address questions/issues in a timely manner.
- **Please be very patient, the support team responds to many inquiries per session**

IS LIVE ATTENDANCE REQUIRED?

- **YES,** if you are taking this training to register for the PMI-ACP exam
- You are **allowed to miss up to two sessions IF** you make up the sessions by **watching the video replays**.
- A **missed** session **means** you were disconnected for **more than a total of 15 mins** for the duration of the session.
- If you **miss three or more sessions**, you will need to make up the missed time by **attending live in another 8-day cohort**.
- *Please see the Bootcamp Calendar for information about upcoming sessions at: http://calendar.skillsoft.com/

ACCESSING THE VIDEO REPLAYS

1. Go to: https://github.com/Skillsoft-Content/PMI-ACP-Replay
2. Replays will be available within 2 business days after the session ends.
3. Click on the Excel file for the year you attended the Bootcamp. You won't see a *file open* option, but it is selected.
4. Click the *Download raw file* button on the far left-hand side.
5. Open the downloaded file using this password: acpB00tcampReplay!

Those are zero's not the letter O. The password is case sensitive.

7. Locate and open the worksheet tab that corresponds with the bootcamp you attended
8. Make a note of the passcode.
9. Paste the provided link into your browser.
10. Complete the required registration steps
11. Input the passcode when prompted
    The password to open the Excel file
    is NOT the passcode to access the replay.

Note: Replays will be available for 1 year.
They are not available for download.

No limit to watch replays to study

PMI®-Authorized PMI Agile Certified Practitioner (PMI-ACP)® Exam Preparation Course

Lesson Four

Delivery

Version 1.0 | 2024 Release

During this segment

- Seek early feedback
- Manage agile metrics
- Manage impediments and risk
- Recognize and eliminate waste
- Perform continuous improvements
- Engage customers
- Optimize flow

Determine which metrics are appropriate for a given audience

Radiate metrics across the relevant audience

Review and analyze metrics

Use metrics insights for decision-making

Seek early feedback

Manage agile metrics

Manage impediments and risk

Recognize and eliminate waste

Perform continuous improvements

Engage customers

Optimize flow

Use metrics insights for decision-making

Section 4 of 4

Velocity

A measure of the average amount of work a team can complete in one iteration

Velocity trends

Project schedule

Potential issues

Realistic goals

Improve team performance

Challenges with velocity

Easy to manipulate

Not easily compared

Internal use

Use CSAT scores to improve customer satisfaction

Identify areas where customers are not satisfied with the product or service being delivered

Use CSAT scores to evaluate team performance

Evaluate the team's performance in delivering a product or service that meets customer needs

Defect density

The number of defects found during a period of development divided by the size of the overall project

Using defect density to help make decisions

Identifying areas for improvement

Comparing projects

Quality goals

Evaluating team performance

Eliminating bottlenecks

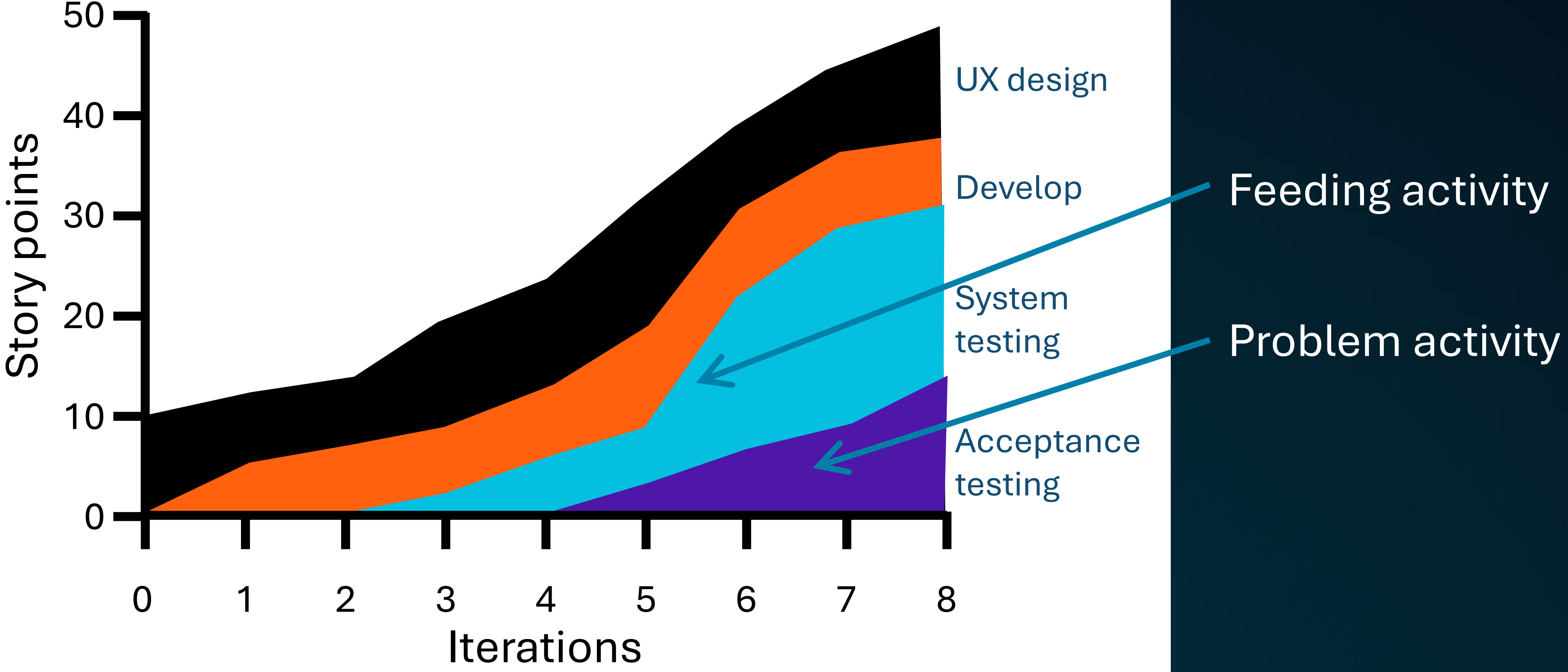
When people performing a specific step in the development process can’t keep up with the amount of incoming work

Spotting bottlenecks

Cumulative flow diagrams

Value stream mapping

Using a cumulative flow diagram to identify a bottleneck



What is a value stream?

A **value stream** is an organizational construct that focuses on the flow of value to customers through the delivery of specific products or services.

Source:
Project Management Institute. (2017). *Agile practice guide*. PMI.

Value stream: Buying a cup of coffee

The customer enjoys the drink

Steps in making a value stream map

Identify the actions

Time worked on

Time waiting

Time between steps

Loopbacks

Visual

You can do the same thing at your organization

Value stream maps make the work taking place visible

This makes it possible for people to have intelligent conversations

Knowledge check

True or false: The value stream begins and ends with the organization.

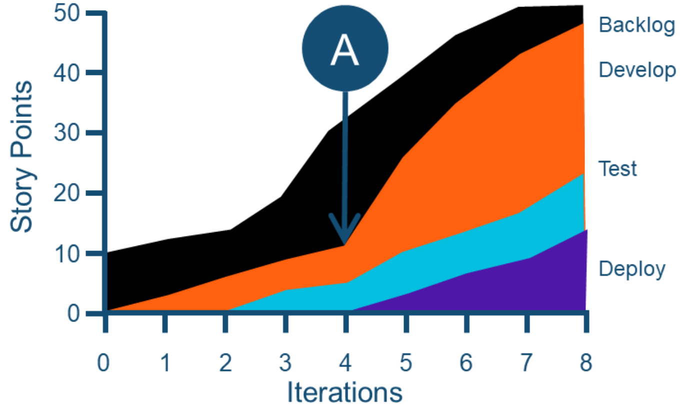
True

False

Knowledge check

What might the inflection at point A in the cumulative flow diagram shown below indicate?

- Excess capacity in test
- **A bottleneck in test**
- A bottleneck in deploy



The sharp increase in the amount of work in Develop could indicate a bottleneck in Test. It appears Test is falling behind since it is the activity below the widening area.

Proactively identify risks and impediments

Engage the team to find the most appropriate course of action

Prioritize impediment removal and risk mitigation activities

Monitor and control risks and impediments

Use lessons learned to avoid risk and impediment recurrence

- Seek early feedback
- Manage agile metrics
- Manage impediments and risk
- Recognize and eliminate waste
- Perform continuous improvements
- Engage customers
- Optimize flow

Proactively identify risks and impediments

Section 1 of 5

What is risk?

Risk is an uncertain event or condition that—if it occurs—will have a positive or negative effect on one or more of the project's objectives.

Source:
Project Management Institute. (2017). *Agile practice guide*. PMI.

Risks can be positive and negative

Risks with positive outcomes

are called opportunities

Risks with negative outcomes

are called threats

What is risk management?

Risk management is the processes of identifying and analyzing project risks and determining appropriate responses.

Agile is not a risk management approach

Building something iteratively does not ensure risks are managed

Layering risk management into agile

Agile approaches offer many cycles and mechanisms that can be used for risk management

Identifying risks and impediments

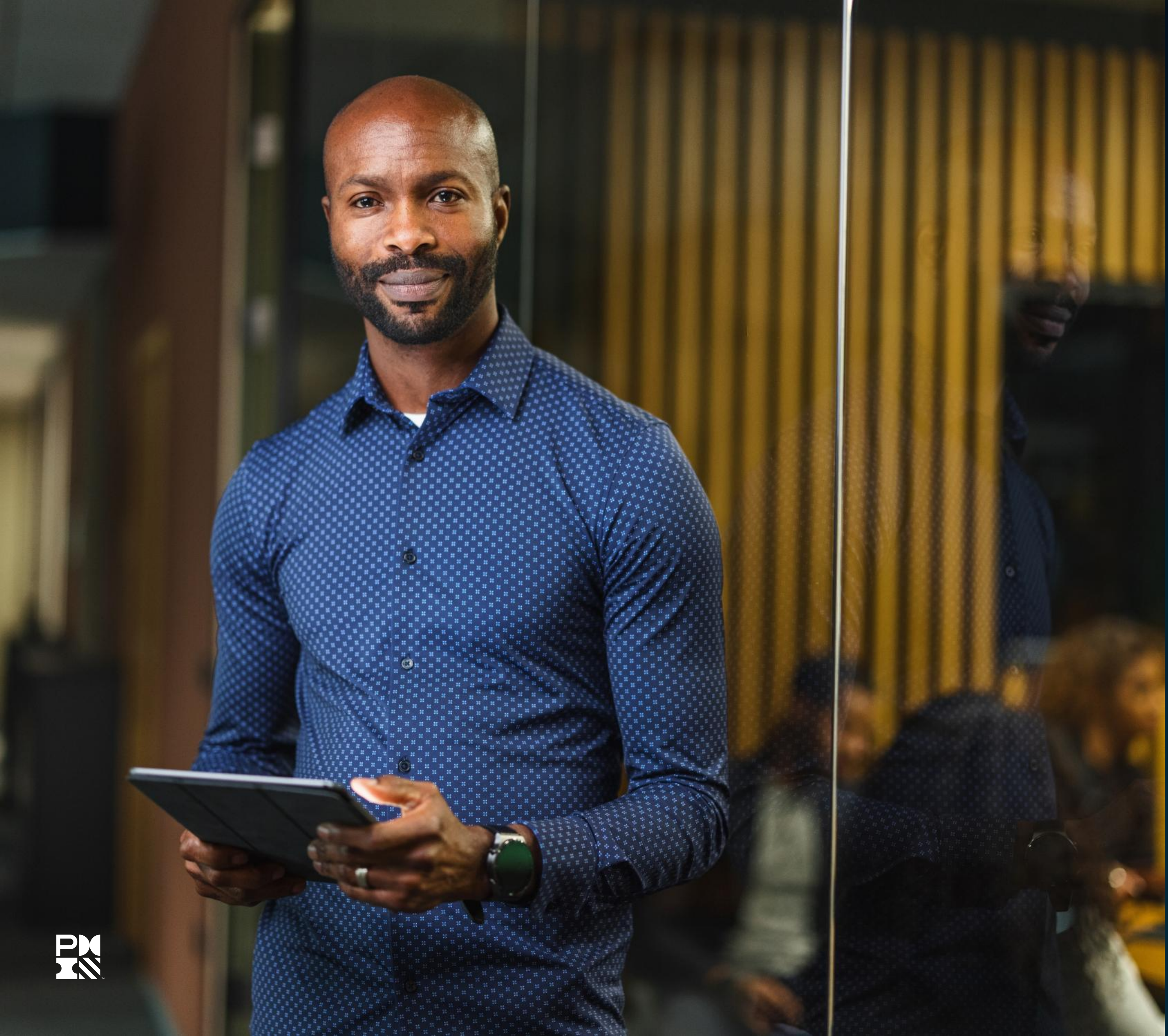
- Collaborative discussions
- Expert judgment
- Formal risk session
- Informal risk session
- Interviews
- Patterns and common lists
- SWOT analysis

Collaborative discussions

Wide range of opinions

Discussion needs to be facilitated

May not be willing to publicly discuss some risks



Expert judgment

Quick

May not have access

Inexperienced teams may ignore risks identified by experts

Formal risk session

Makes risks visible

Useful in disagreement

Regulatory compliance

Advance preparation

Difficult to schedule

Rubber-stamping

Informal risk session

Inclusive

Minimal preparation

Difficult to schedule

Applicable to regulatory situations

Interviews

Private discussions about risks that people may not be willing to discuss openly

Miss risks when not discussed as a group

Patterns/ common lists

Reusing lists

New types of risks may be missed

SWOT analysis

Identifies positive risks

Competitive situations

Collaborative

group discussions

Goes beyond threat

identification

Can take time

Engage the team to find the most appropriate course of action

Section 2 of 5

Why engage the team?

Better decisions

Promotes problem-solving

No endless appeals

Fosters action

Source: Wondolleck, J., & Yaffee, S. (2000). *Making collaboration work: Lessons from innovation in natural resource management*. Island Press.

Collaboration isn't magic

There are downsides and potential for misuse

The benefits far outweigh any potential liabilities

Planning risk responses

Avoid

Reduce

(mitigate)

Transfer

Escalate

Accept

Best ⟷ Worst

Business value retained

Denial

Not a valid response

Can be catastrophic

Spike

A spike is a short time interval within a project, usually of fixed length, during which a team conducts research or prototypes an aspect of a solution to prove its viability.

Source: Project Management Institute. (2017). *Agile practice guide*. PMI.

Spikes to explore and reduce risk

Spikes typically take less time than a standard iteration

Short iterations

Allow us to check frequently and ensure our risk responses are working

Knowledge check

True or false: Denial is a valid response to risk

True

False

Knowledge check

True or false: Avoiding risk is preferable to accepting it

True

False

Prioritize impediment removal and risk mitigation activities

Section 3 of 5

Risk-adjusted backlog

Risk response activities

Collaborative discussion

Reduces amount of risk

Source:

[2.3 Assess and Manage Risks - PM Illustrated PMP Exam]

Put risk responses into the team's backlog

Make sure risk responses are a regular part of the team's day-to-day work

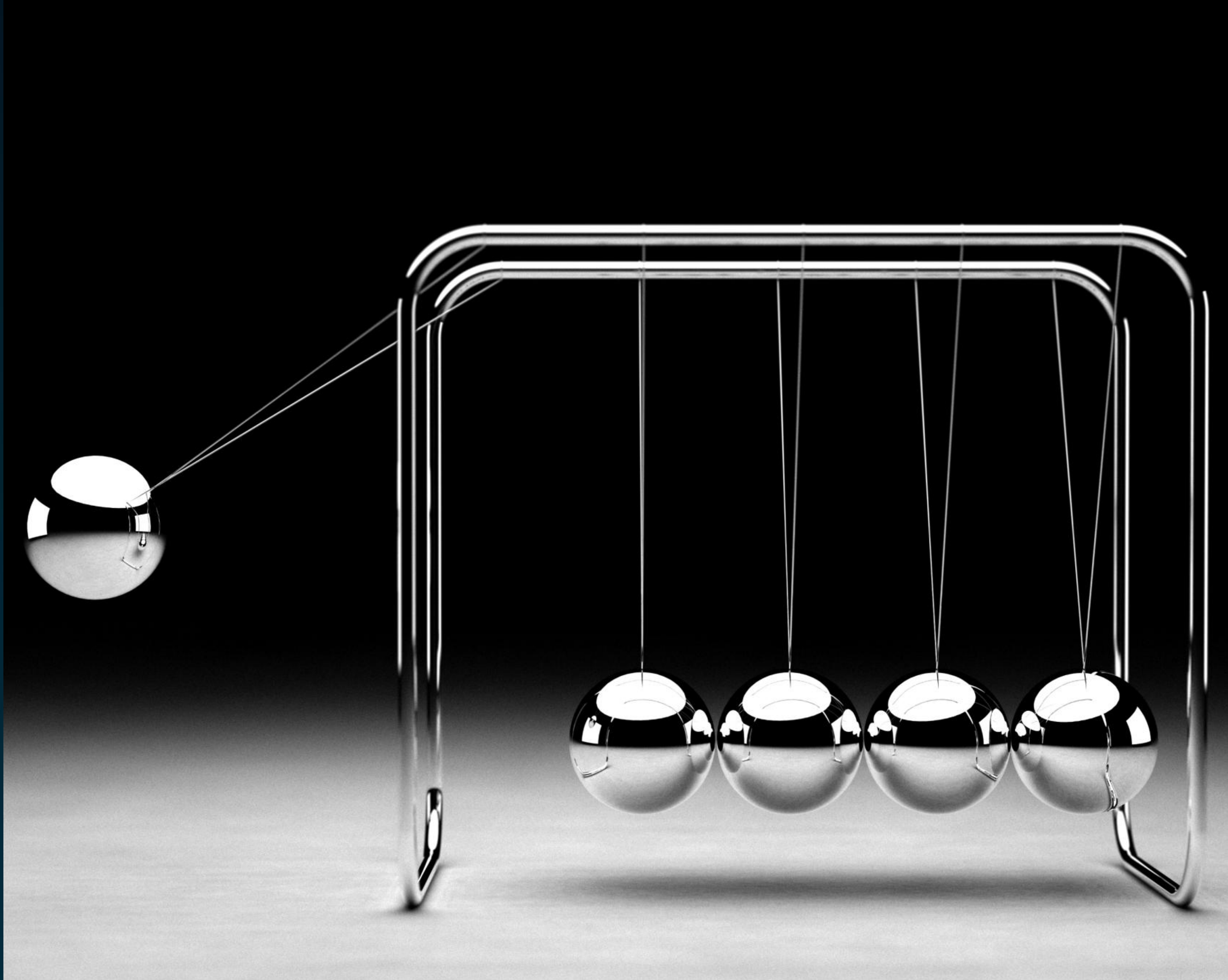
Residual risk

Risk that remains after your risk responses have been implemented

Should be clearly identified, analyzed, documented, and communicated

Secondary risks

How do we account for the effect secondary risks have on the primary risks that triggered them?



Adjusting the value of a risk response action

Net EMV = EMV of residual risk + EMV of secondary risk

Risk with EMV of US$60,000

Response only addresses half the problem

Secondary risk with EMV of US$10,000

US$60,000 X 50% = US$30,000

US$30,000 + US$10,000 = US$40,000

Add to the risk-adjusted backlog

Feature	US$50,000
Feature	US$40,000
Feature	US$30,000
Feature	US$20,000
Risk response	US$20,000
Feature	US$5,000

Repeat the process for each of the remaining risk responses

Work with the product owner to create the risk-adjusted backlog

Product owners rank the work based on its business value

That includes work undertaken to avoid, reduce, escalate, or transfer risk

Knowledge check

How do we prioritize impediment removal and risk mitigation activities ?

A. Discuss them in daily coordination meetings

B. Reach out to the product owner

C. **Insert them into the backlog**

D. Poll relevant stakeholders

The correct answer is insert them into the backlog. By doing so, we make sure our risk responses are not supplemental activities but a regular part of the team's day-to-day work.

Knowledge check

How do we account for the effect secondary risks have on the primary risks that triggered them?

A. Add the response to the risk-adjusted backlog

B. Estimate the net expected monetary value

C. Work with the product owner

D. Identify, analyze, document, and communicate residual risk

The correct answer is estimate the net expected monetary value.

Monitor/control risks and impediments

Section 4 of 5

Risk retrospectives

A team review of the project risks identified and risk management processes used

How does the team use this meeting?

Still active?

Changes to probability, impact, or urgency

Effectiveness of risk responses

Identify new risks; close those no longer active

Risk retrospective questions

98

Are we eliminating or reducing our risks?

Do we have any new or increasing risks?

What are the root causes of our risks, and can we eliminate any of them?

Which risk strategies are working, and which are not?

For risks that we chose to transfer or escalate,
how are the third parties managing them?

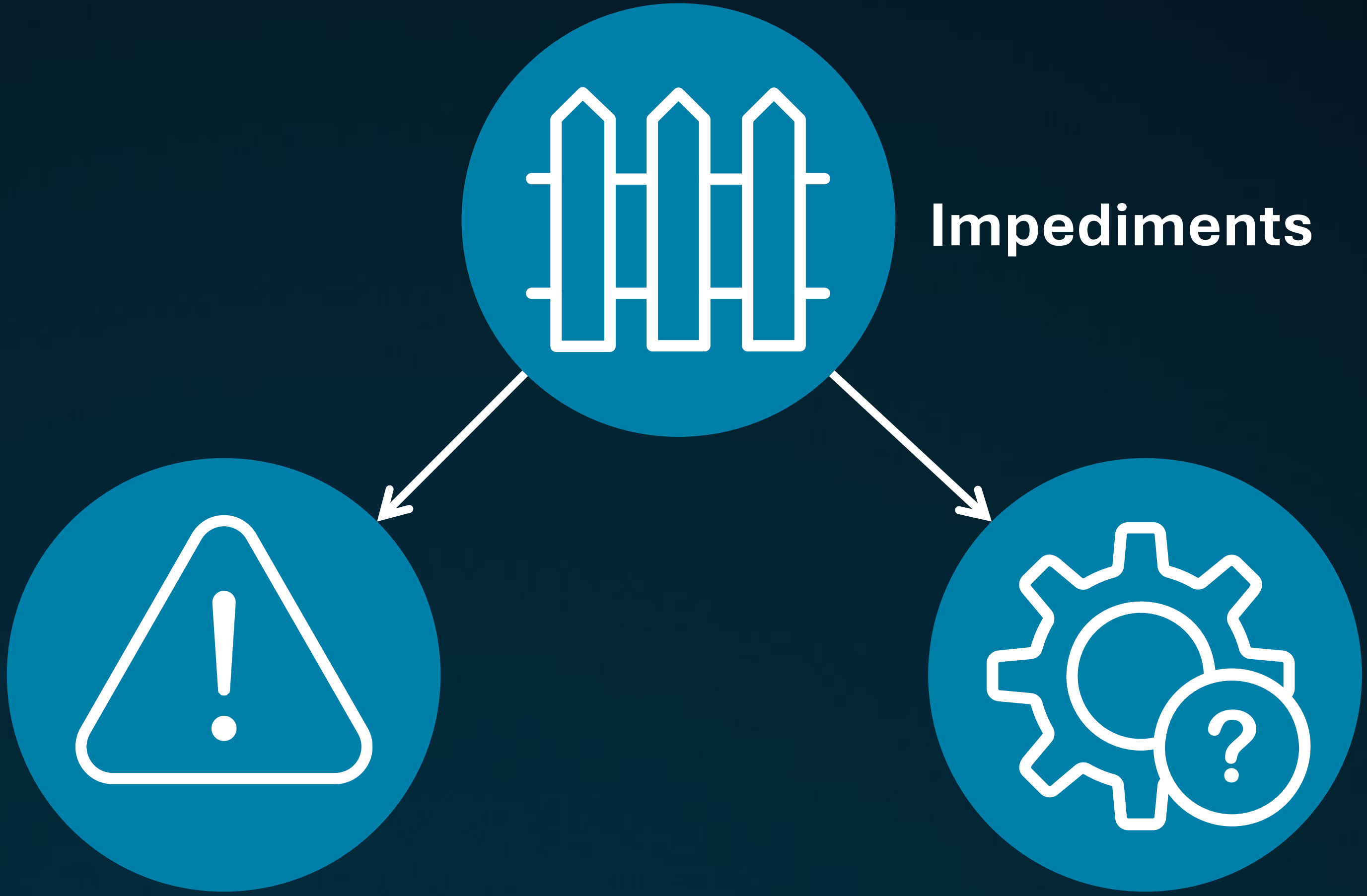
How are our team risk management capabilities developing?

Reprioritize the backlog

- Feature
- Risk response
- Feature
- Feature
- Risk response
- Feature

The product owner should use the results of your risk retrospective to reprioritize the backlog

Impediments raised at daily coordination meetings



Impediments

Risks

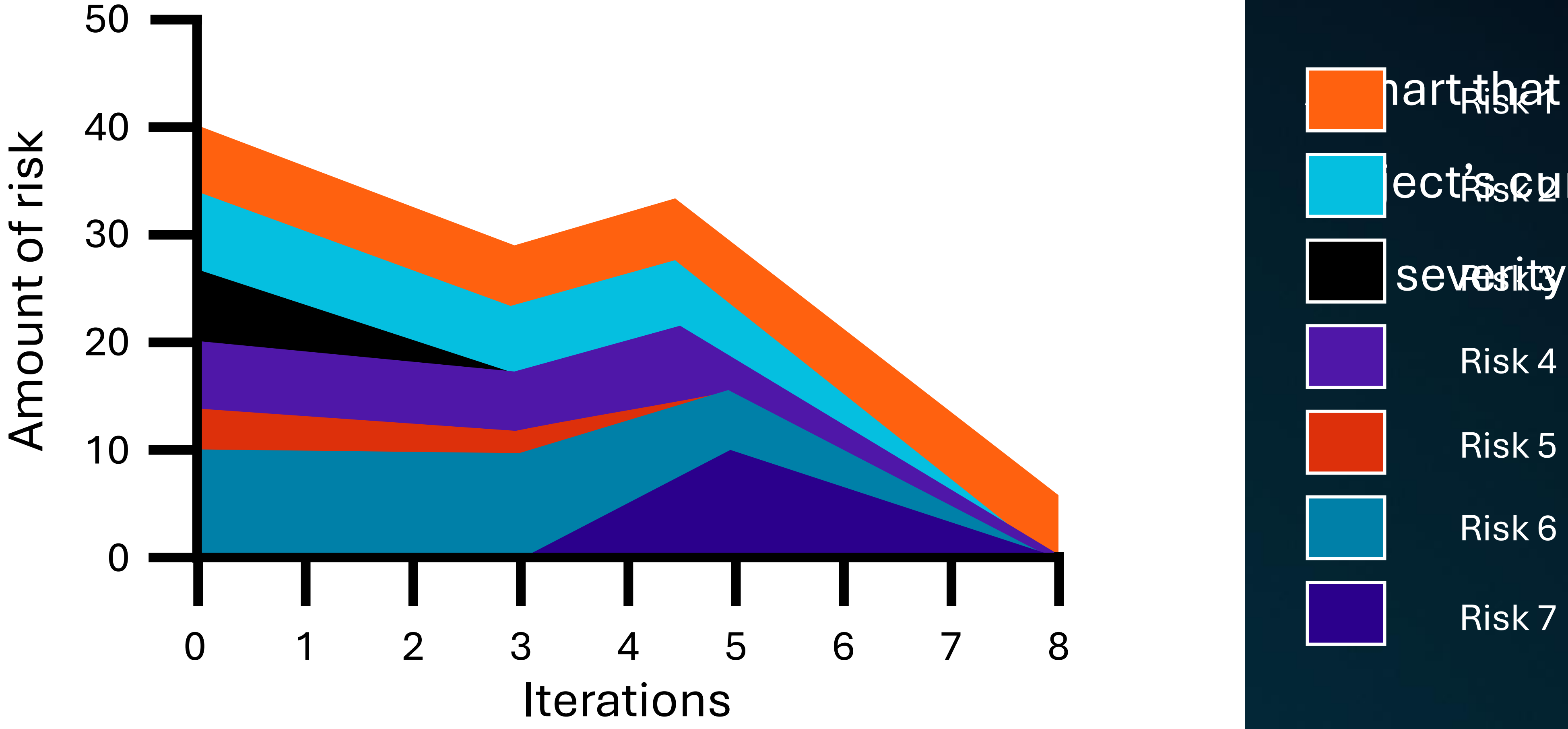
(have not yet occurred)

Issues

(have occurred)

Visualizing risks

One tool that can help make sense of all this data is a risk burndown graph



What is a risk burndown graph?

…hart that shows the …ject's cumulative severity

Risk management should be a team activity

We want risk management activities to be part of the day-to-day activities of a larger pool of project stakeholders

Knowledge check

True or false: Issues are risks that have already happened

True

False

Knowledge check

True or false: Risk retrospectives should be held at the end of the project

True

False

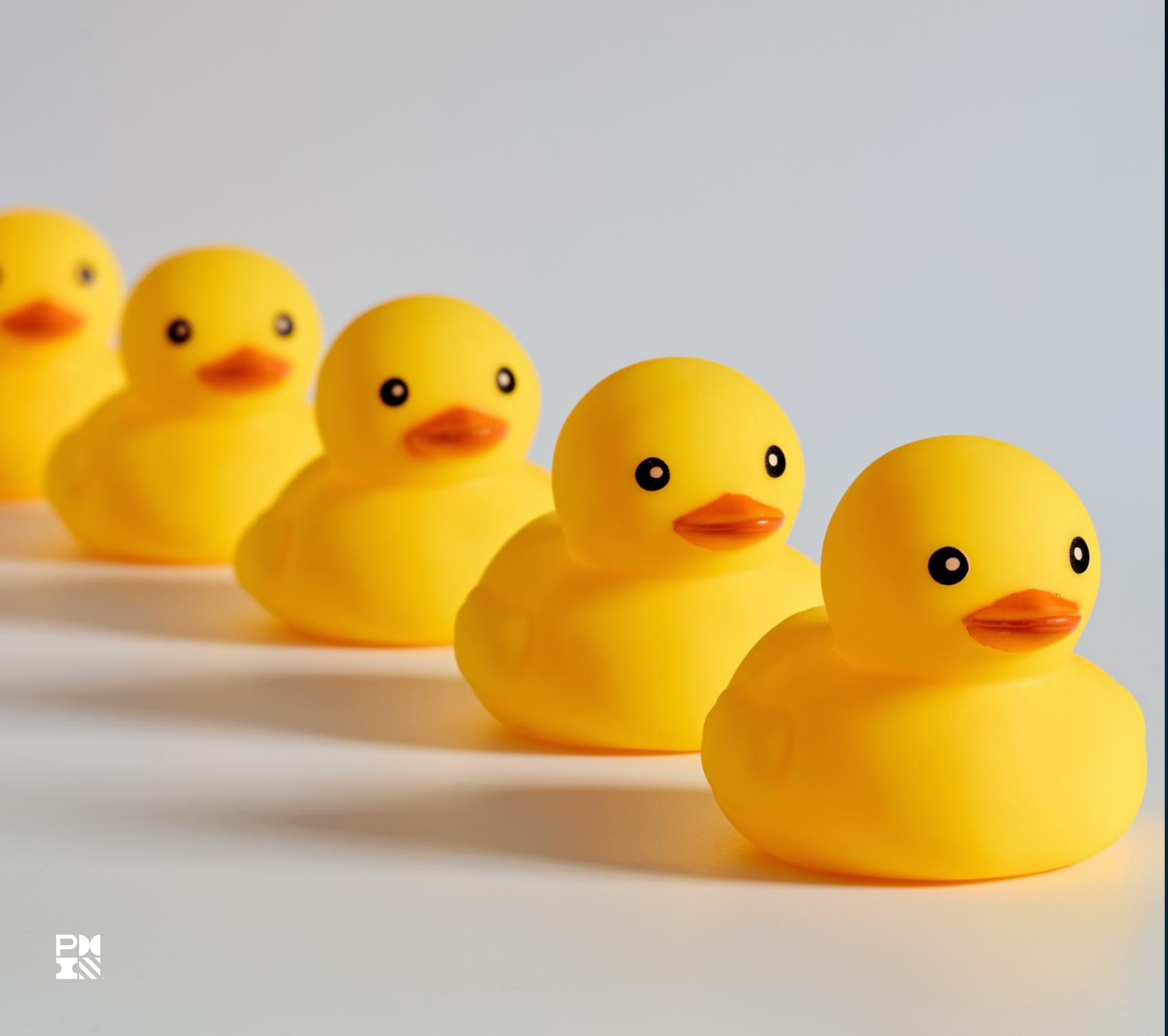
Use lessons learned to avoid risks/impediments recurrence

Section 5 of 5

Lessons learned

Lessons learned is the knowledge gained during a project, which shows how project events were addressed or should be addressed in the future, for the purpose of improving future performance.

Source: Project Management Institute. (2023). *A Guide to the Project Management Body of Knowledge* (PMBOK® Guide). Project Management Institute.



Use lessons learned to avoid recurrence

If another team has faced and overcome a similar risk in the past, we may be able to use their strategy to overcome our own risks

Review lessons learned from similar and parallel projects

Past similar or parallel projects may have solved risks your project faces

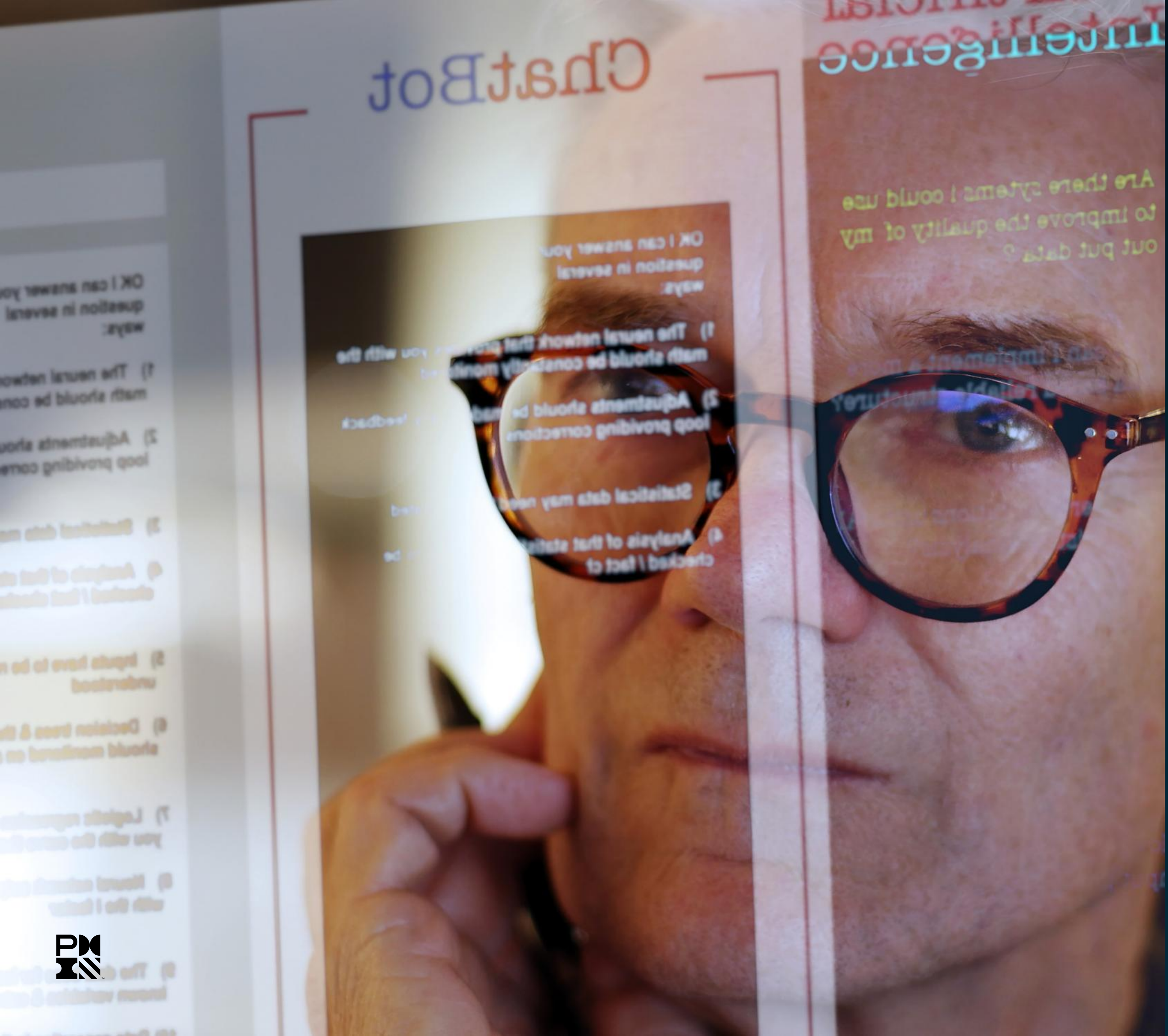
Ask team for solutions

Their responses may provide a sort of tacit lessons learned

Use communities of practice

Comes together to learn from each other and develop their skills and knowledge

Can be ideal places to share and understand lessons learned



Artificial intelligence

You can ask an AI chatbot to find potential risks and lessons learned that are pertinent to your domain

Be careful about the information you share with a chatbot

Knowledge check

What are lessons learned?

A. Problems from prior projects

B. Problems faced by another team

C. A way to develop your team's skills and knowledge

D. The knowledge gained during a project

Lessons learned are the knowledge gained during a project, which shows how project events were addressed or should be addressed in the future, for the purpose of improving future performance.

Visualize the end-to-end flow of value in the system

Use metrics, tools, and feedback loops to identify waste

Prioritize waste-reduction activities

Iterate on identification and reduction of waste

Seek early feedback

Manage agile metrics

Manage impediments and risk

Recognize and eliminate waste

Perform continuous improvements

Engage customers

Optimize flow

Visualize the end-to-end flow of value in the system

Section 1 of 4

Definitions

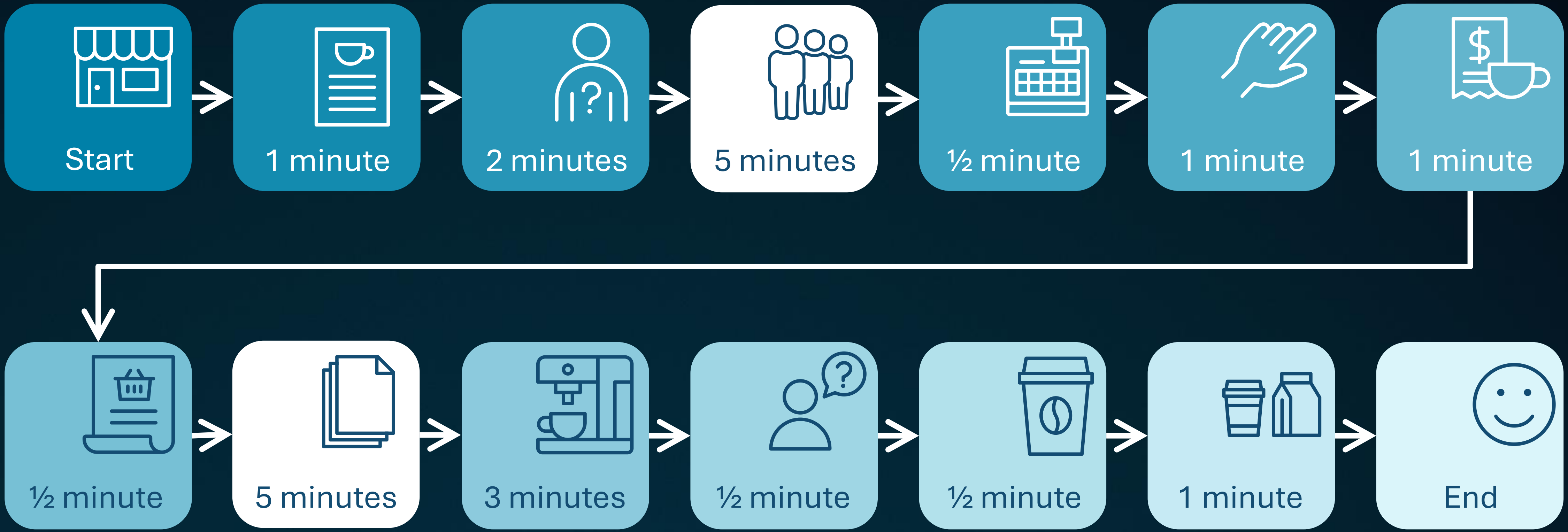
Value-added time is the time during the cycle that value is being added.

Non-value-added time is all the rest of the cycle, during which we find delays, waste, and constraints.

Total cycle time is the sum of a process's value-added and nonvalue-added times.

Identify steps and queues in the process

Estimate time for each step



Value-added or non-value-added?

Process efficiency

Process efficiency = Value-added time ÷ Total cycle time

Value-added time:	**10 minutes**
Non-value-added time:	**11 minutes**
Total cycle time:	**21 minutes**

10 minutes ÷ 21 minutes = 48%

Optimizing the process

We need to reduce or,

if possible, eliminate

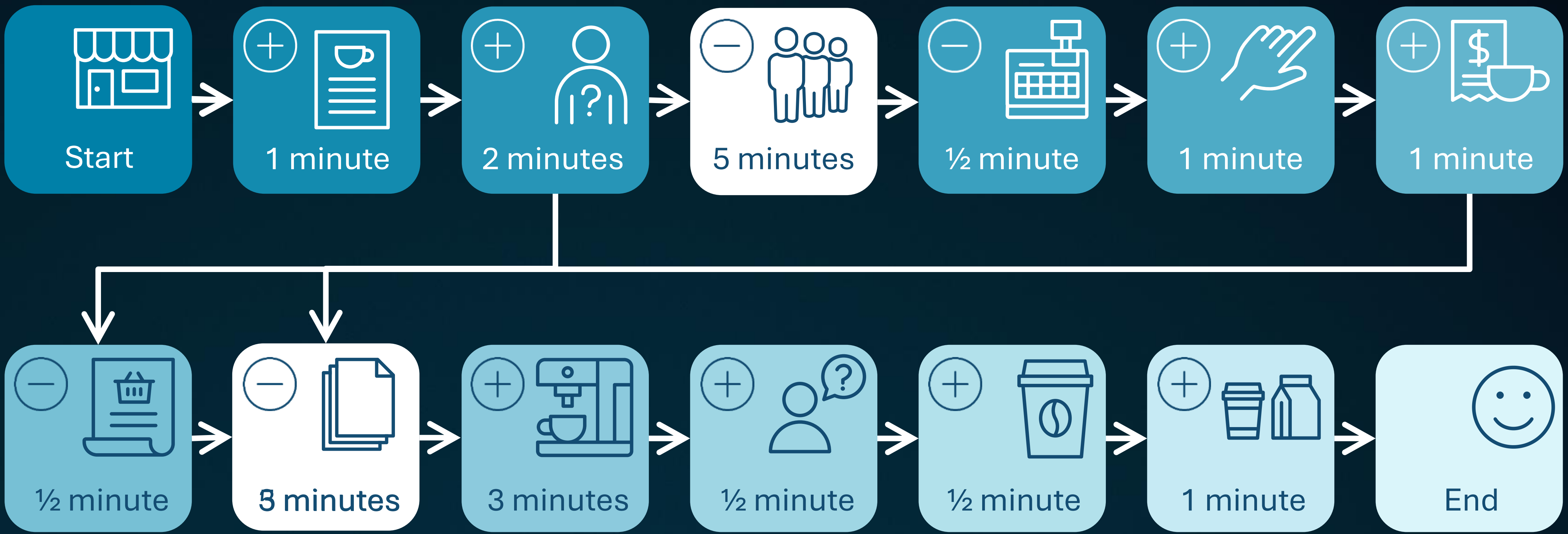
wait times and loopbacks

Developing an app

This won't eliminate people wandering into the coffee shop and ordering something in person

It should significantly reduce the wait to order and pay for a drink

App value stream



App process efficiency

Value-added time:	8 minutes
Non-value-added time:	3 minutes
Total cycle time:	11 minutes

8 minutes ÷ 11 minutes = 73%

Use metrics, tools, and feedback loops to identify waste

Section 2 of 4

Metrics considerations

Fit for purpose

Specific

Aligned

Consistent

Actionable

Timely

Useful metrics categories

Queue length

Work in progress

Throughput

Process efficiency

Queue length

Can provide insight into the amount of work in progress and the rate of work completion

- Cycle time
- Lead time
- Queue time
- Batch size

Work in progress

The amount of work currently being worked on that has not been completed

- Estimate at completion (EAC)
- Ratio of story points completed to the total story points planned

Throughput

The rate at which the team can complete work items over a given period

- Features complete versus features remaining
- Velocity

Process efficiency

The team's ability to deliver value to the customer while minimizing waste and maximizing productivity

- Cycle time
- Lead time
- Velocity

Tools

Value stream mapping

Theory of constraints

Cumulative flow diagram analysis

Value stream map analysis

Theory of constraints

Throughput can only be improved when the constraint is improved

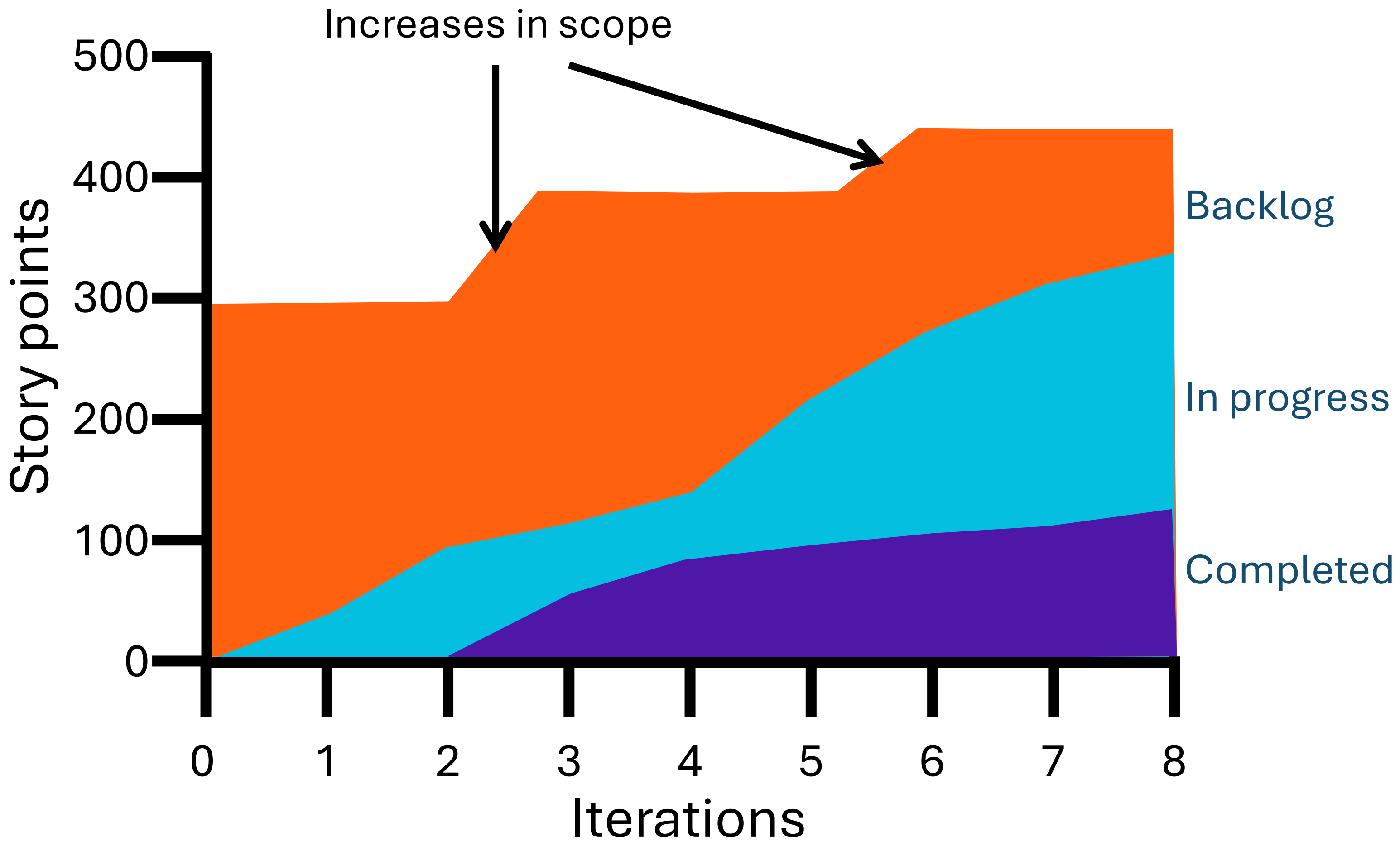
Spending time optimizing nonconstraints will not provide significant benefits

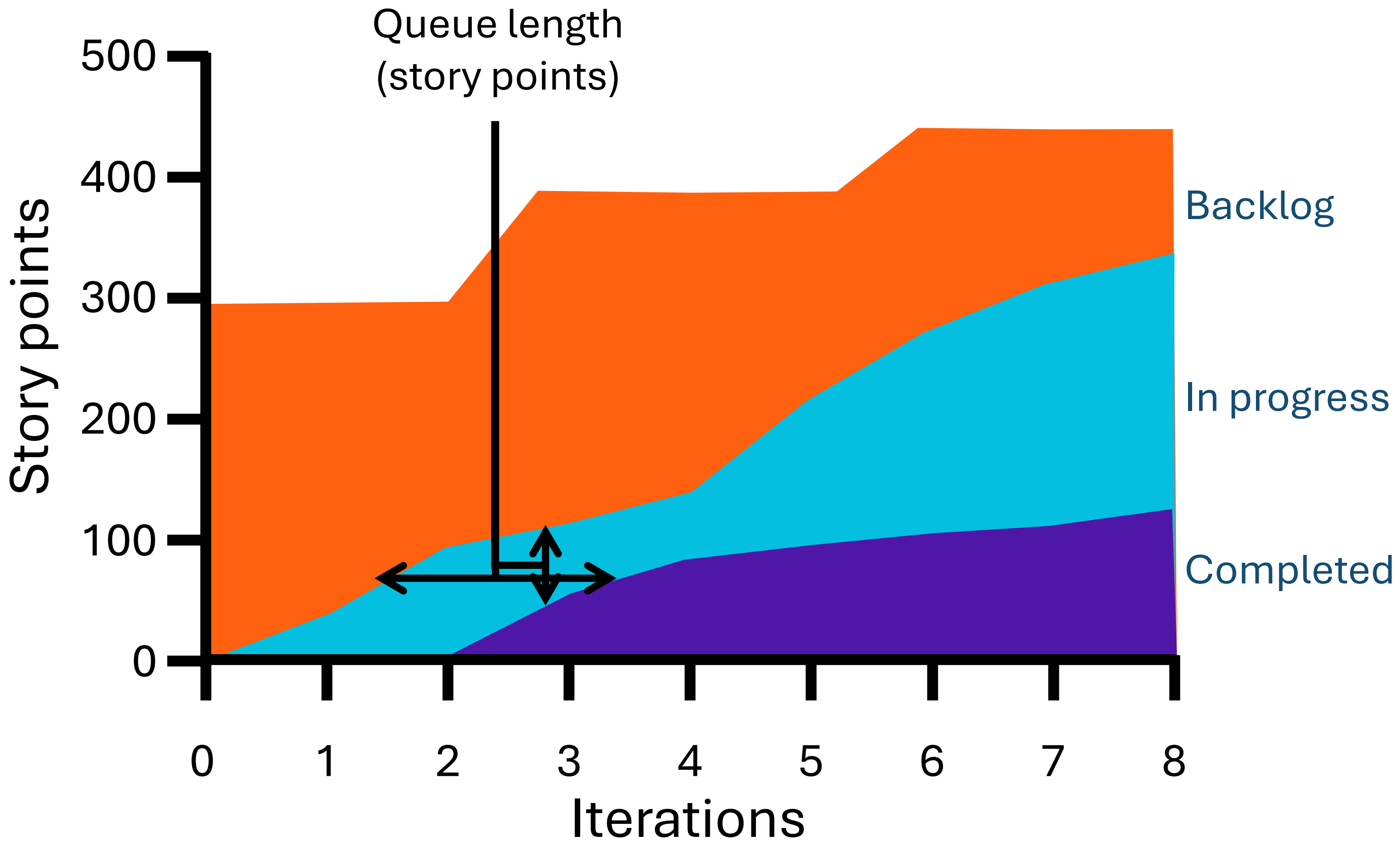
Sources:

Goldratt, E. M., & Cox, J. (1984). *The goal*. North River Press, Inc.
Goldratt, E. M. (2017). Critical *chain*. Routledge.

Cumulative flow diagrams:

Increases in scope





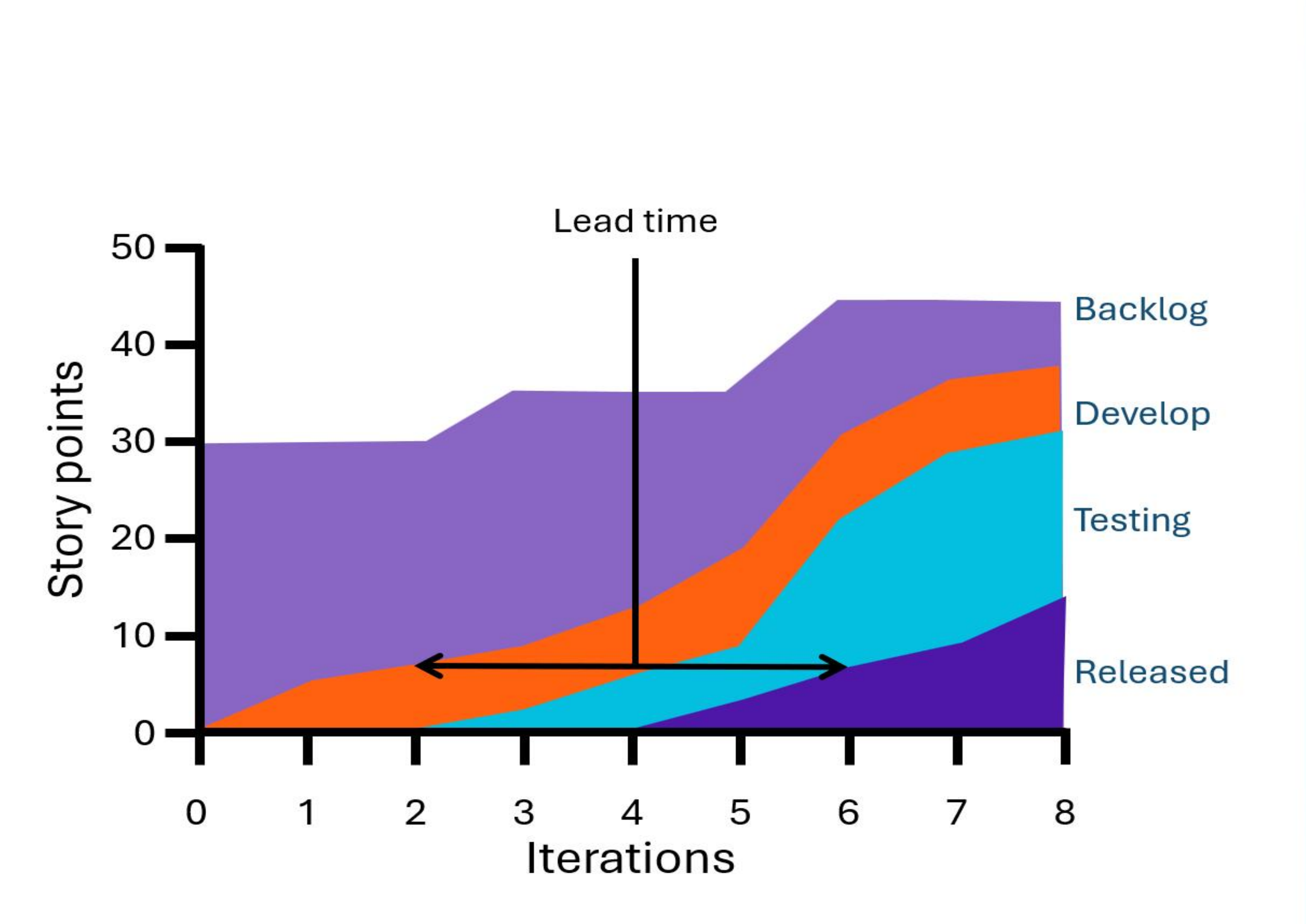
Cumulative flow diagrams:

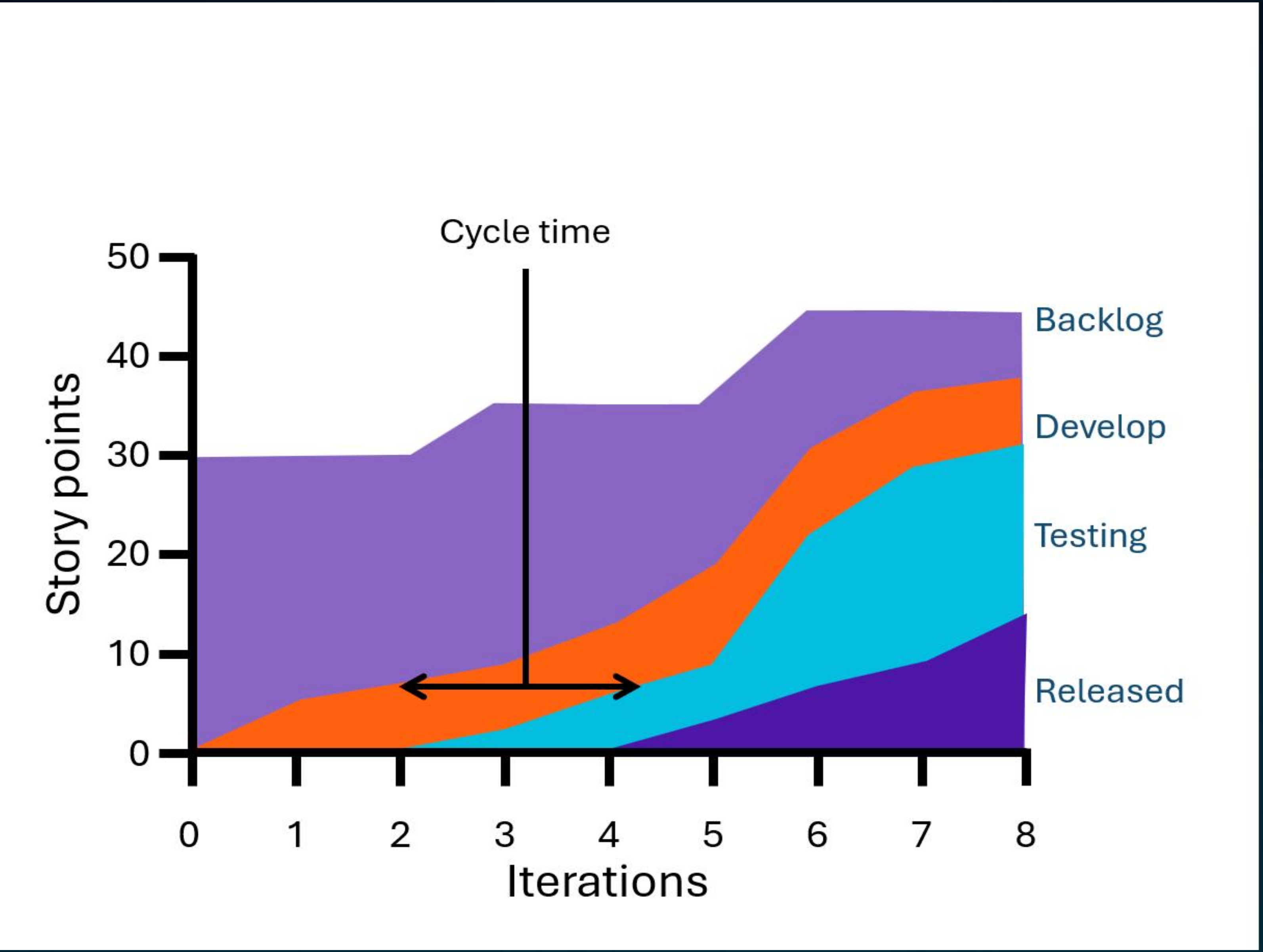
Queue duration and length

Little's Law: A mathematical formula from queuing theory that proves the duration of a work queue is dependent on its size

Cumulative flow diagrams:

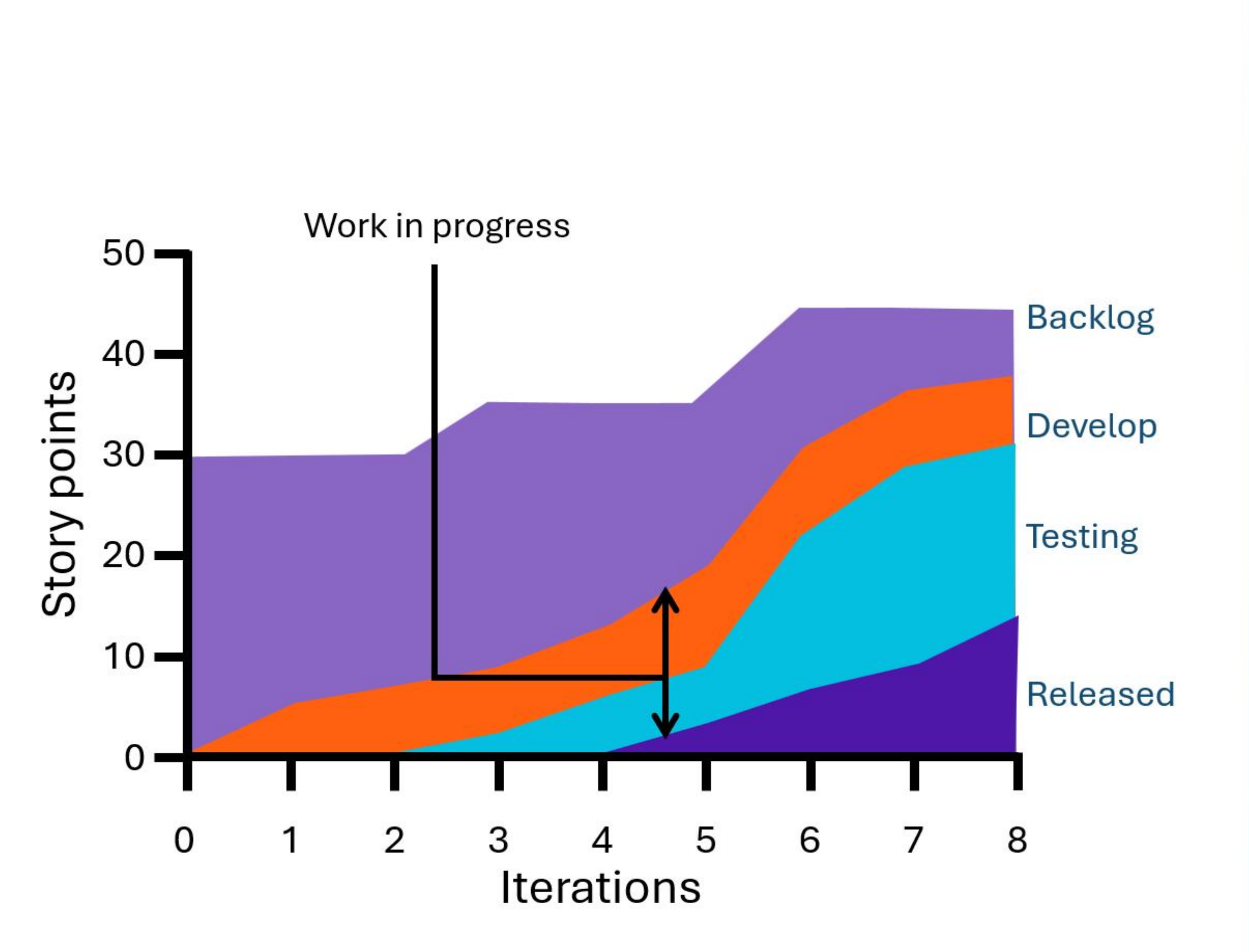
Lead time, cycle time, and work in progress





Cumulative flow diagrams:

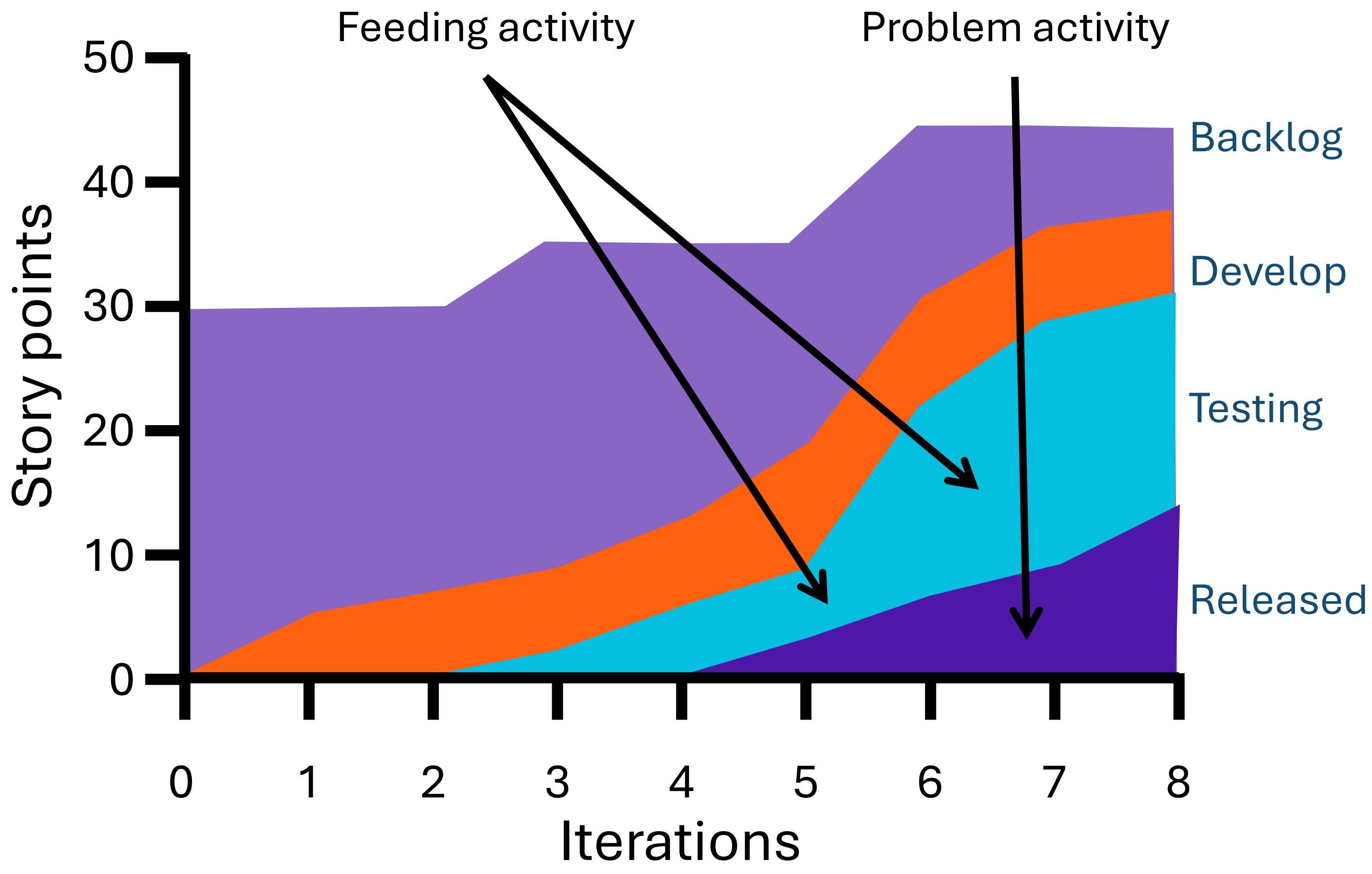
Lead time, cycle time, and work in progress



Cumulative flow diagrams:

Lead time, cycle time, and work in progress

Using a cumulative flow diagram to identify a bottleneck



Knowledge check

True or false: Queue length can provide insight into the amount of work in progress and the rate of work completion

True

False

Knowledge check

True or false: According to the theory of constraints, spending time optimizing nonconstraints can still provide significant benefits

True

False

Prioritize waste-reduction activities

Section 3 of 4

Putting waste-reduction tasks in the backlog

Feature

Waste-reduction task

Feature

Feature

Waste-reduction task

Feature

How do we put a value on waste-reduction tasks so the product owner can prioritize them?

Estimating the value of waste-reduction tasks

Determine the total value of the waste

Calculate the potential savings

Calculate the return on investment

Prioritize the task

Work with the product owner to set priorities

You may need to explain why waste-reduction activities are important

Team selects and works on waste-reduction activities

Review the results

If it was effective, the team moves on to the next task

If it was not, more work may be necessary

Knowledge check

How do you estimate the value of a waste-reduction task?

A. Calculate the return on investment, determine the total value of the waste, calculate the potential savings, prioritize the task, calculate the return on investment

B. Determine the total value of the waste, calculate the potential savings, calculate the return on investment, prioritize the task

C. Calculate the potential savings, prioritize the task, calculate the return on investment, determine the total value of the waste

To estimate the value of a waste-reduction task, you first determine the total value of the waste, then calculate the potential savings, then calculate the return on investment, and finally prioritize the task based on the estimated return on investment

Iterate on identification and reduction of waste

Section 4 of 4

Theory of constraints

Provides precise and sustained focus on improving the current constraint

Once the constraint is improved, all that focus moves to the next constraint

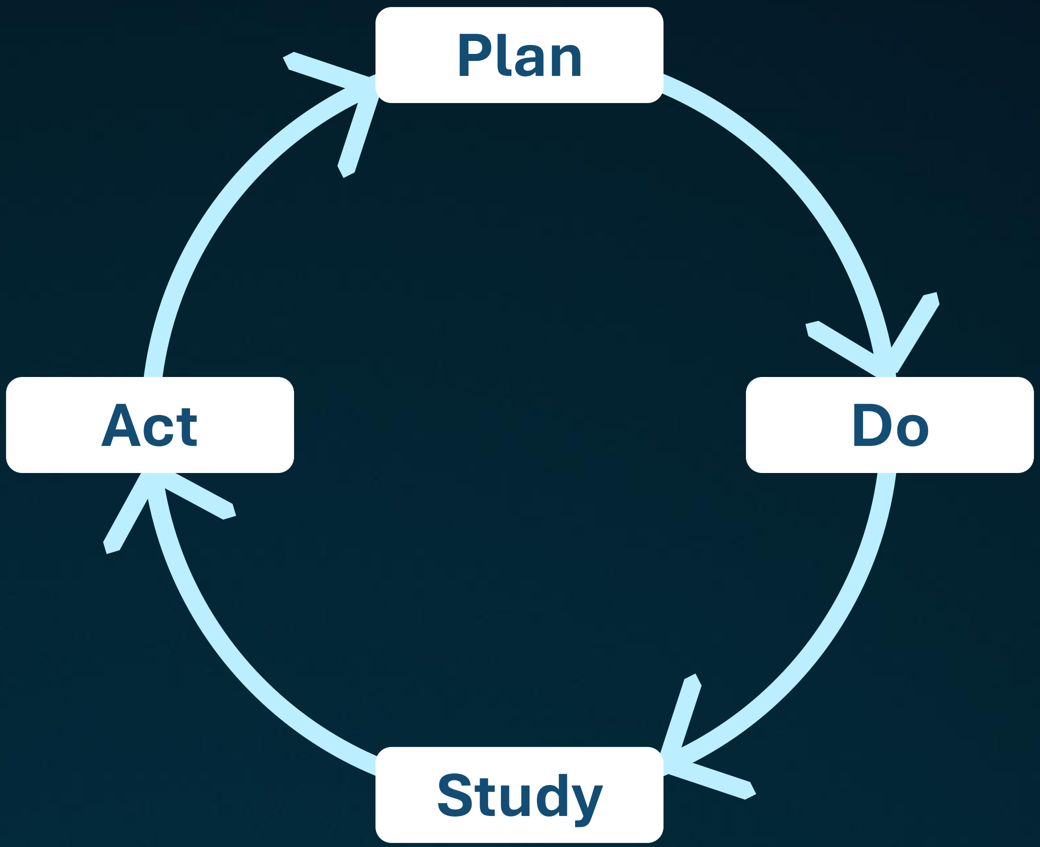
Sources:
Goldratt, E. M., & Cox, J. (1984). *The goal*. North River Press, Inc.
Goldratt, E. M. (2017). Critical *chain*. Routledge.

Quality loop

Expose the next weakest link in the chain, strengthen it, and move to the next one

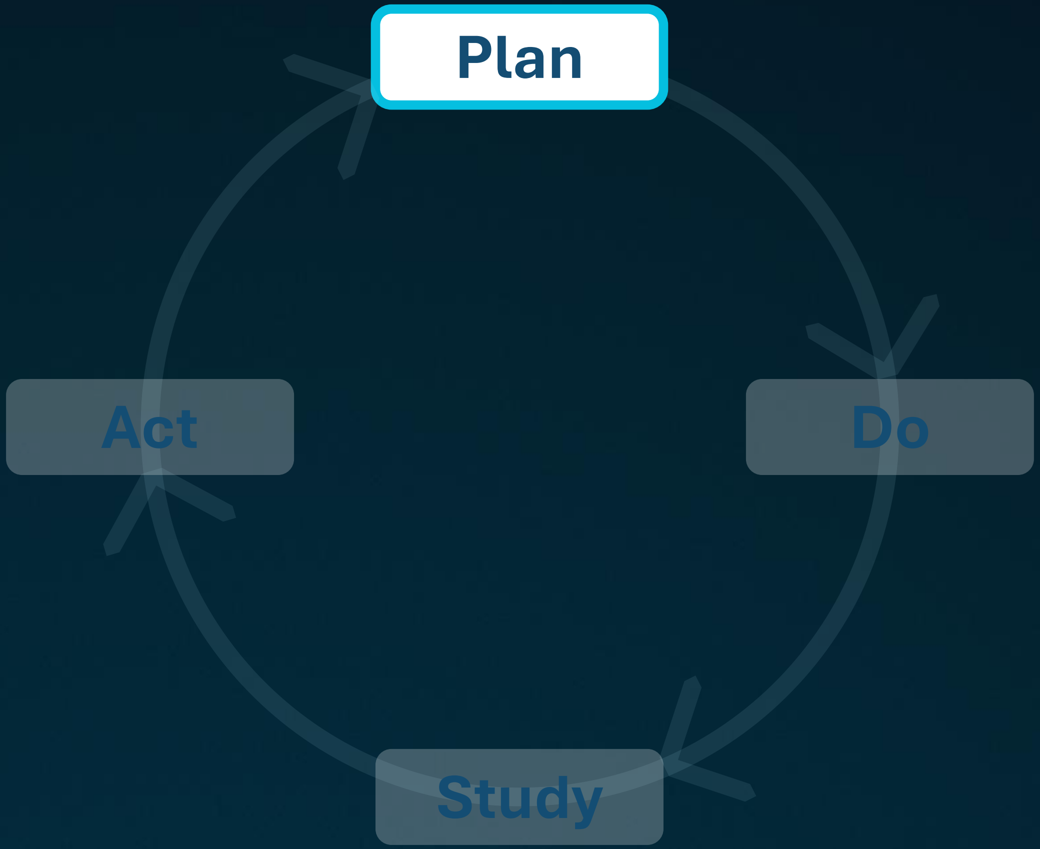
Don’t let inertia become a constraint to improvement

Deming Cycle



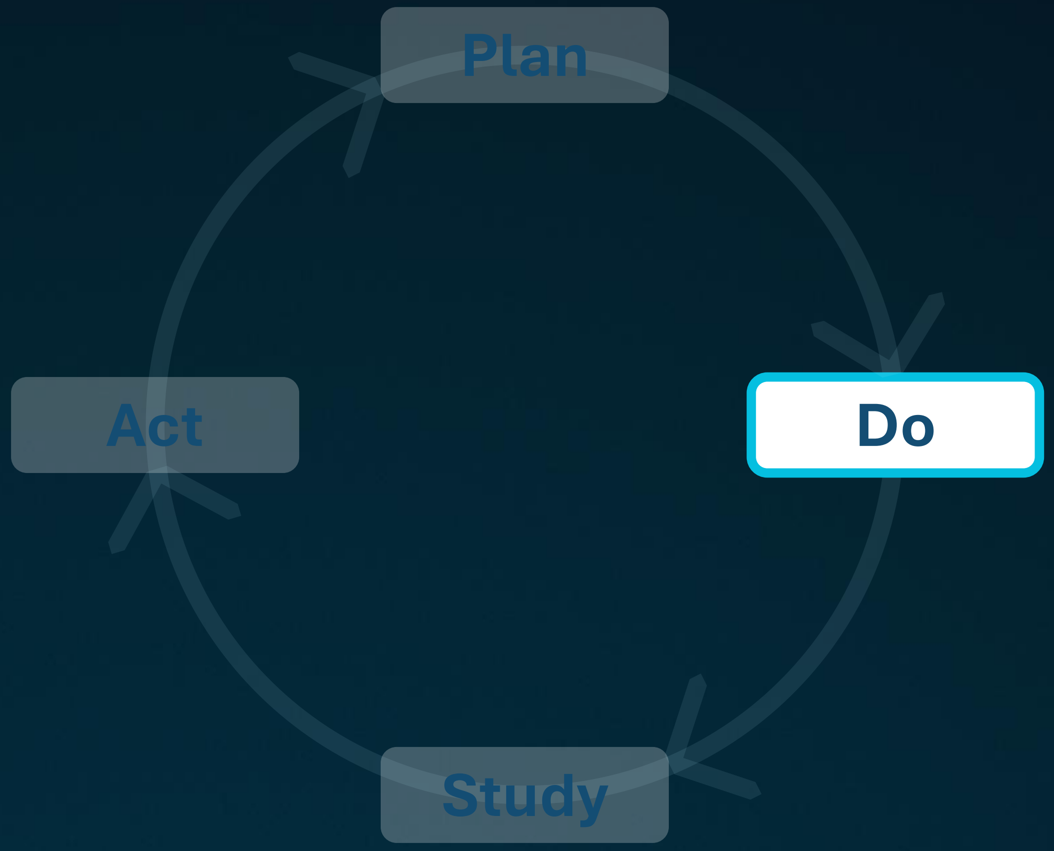
Source: Deming, W. Edwards (1982). *Out of the crisis*. Center for advanced engineering study, Center for Advanced Engineering Study, MIT.

Deming Cycle: Plan



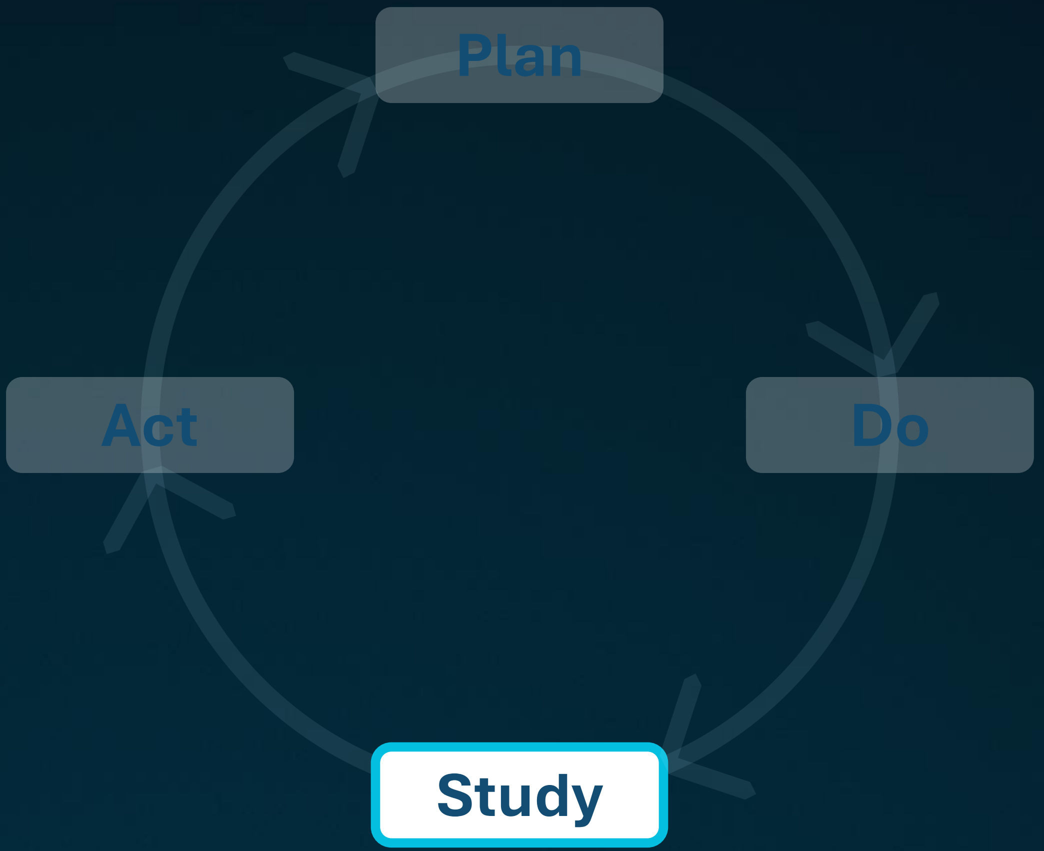
Source: Deming, W. Edwards (1982). *Out of the crisis*. Center for advanced engineering study, Center for Advanced Engineering Study, MIT.

Deming Cycle: Do



Source: Deming, W. Edwards (1982). *Out of the crisis*. Center for advanced engineering study, Center for Advanced Engineering Study, MIT.

Deming Cycle: Study

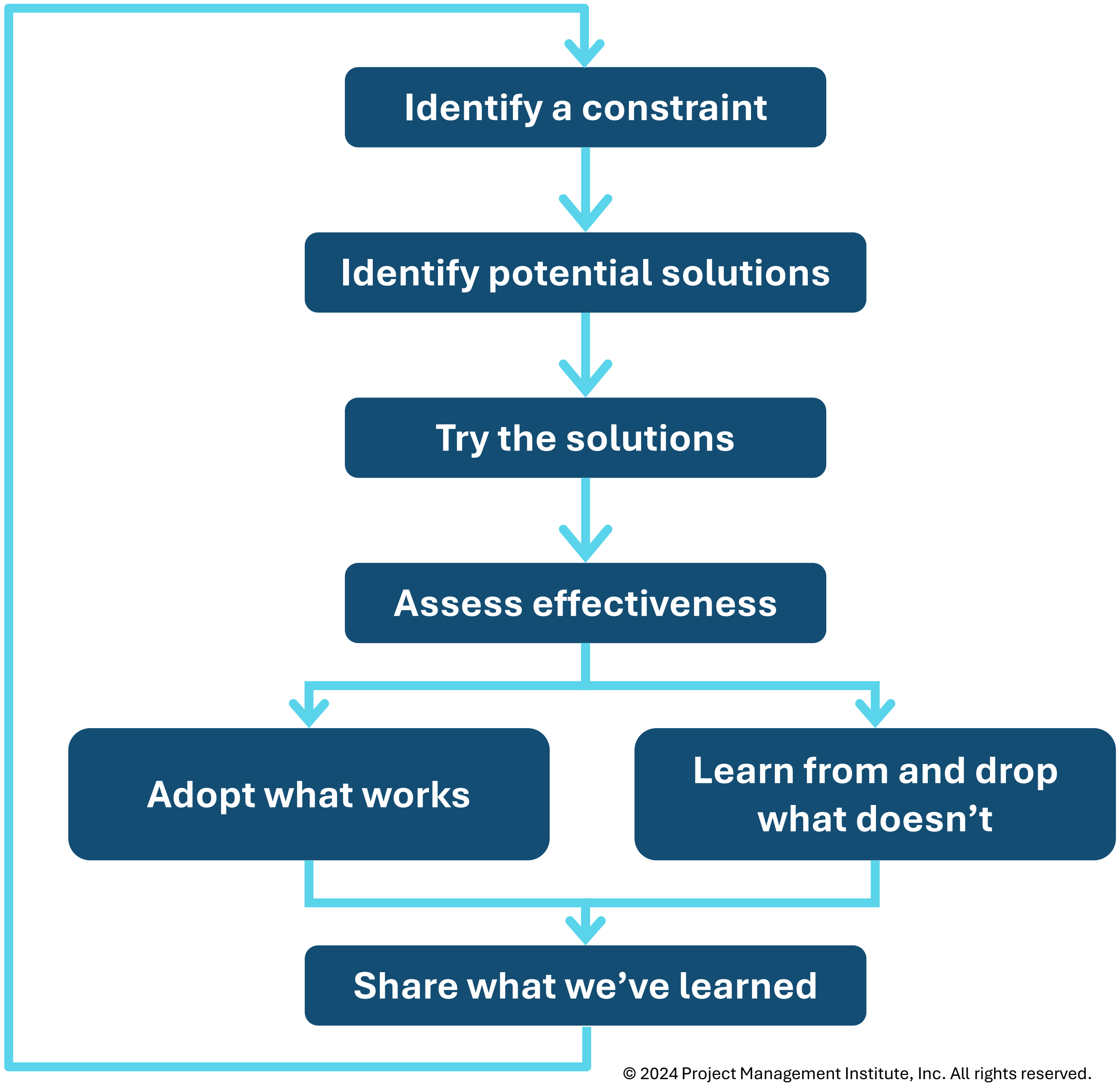


Source: Deming, W. Edwards (1982). *Out of the crisis*. Center for advanced engineering study, Center for Advanced Engineering Study, MIT.

Deming Cycle: Act

Source: Deming, W. Edwards (1982). *Out of the crisis*. Center for advanced engineering study, Center for Advanced Engineering Study, MIT.

Improving constraints through experiments



Knowledge check

True or false: In Deming's PDSA Cycle, the letters stand for prepare, do, study, and act

True

False

Knowledge check

True or false: The theory of constraints provides precise and sustained focus on improving the current constraint

True

False

DAILY PMP BOOTCAMP SURVEY

LOOK FOR THE SURVEY LINK IN THE CHAT

- Our goal is to provide the best possible Bootcamp experience for a live streaming webinar, with hundreds of participants.

- For each Bootcamp session,
  - Let us know **what you liked** about the experience – your comments really matter.
  - Please include a thank you **to the mentor(s)** working off camera.
  - If you have **recommendations**, share those too!

We sincerely value your opinion!

Survey Scale

- This Scale: 0 not at all likely- 10 extremely likely

skillsoft

On a scale of 0-10, how likely are you to recommend this bootcamp to someone else?

This Scale: 0 not at all likely - 10 extremely likely

0	1	2	3	4	5	6	7	8	9	10
○	○	○	○	○	○	○	○	○	○	○